Study and Implementation of **Contrast Limited Adaptive Local Histogram Equalization Method for Poor Contrast Image Enhancement**

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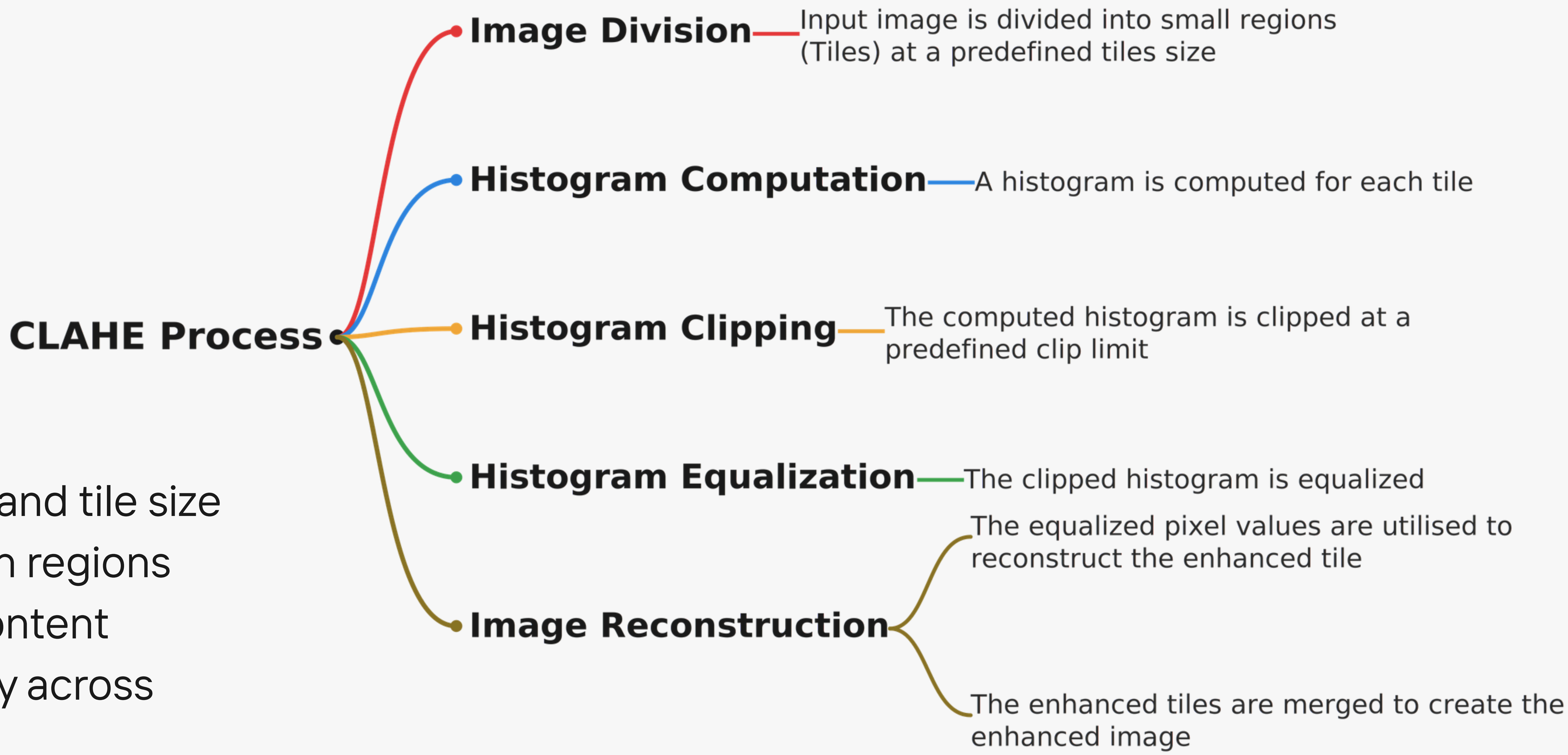
The Problem With CLAHE



- Manual tuning of clip limit and tile size
- Leads to artifacts in certain regions
- No adaptation to image content
- Hurts enhancement quality across diverse images

Claims of CLALHE

1

Fully automatic parameter selection

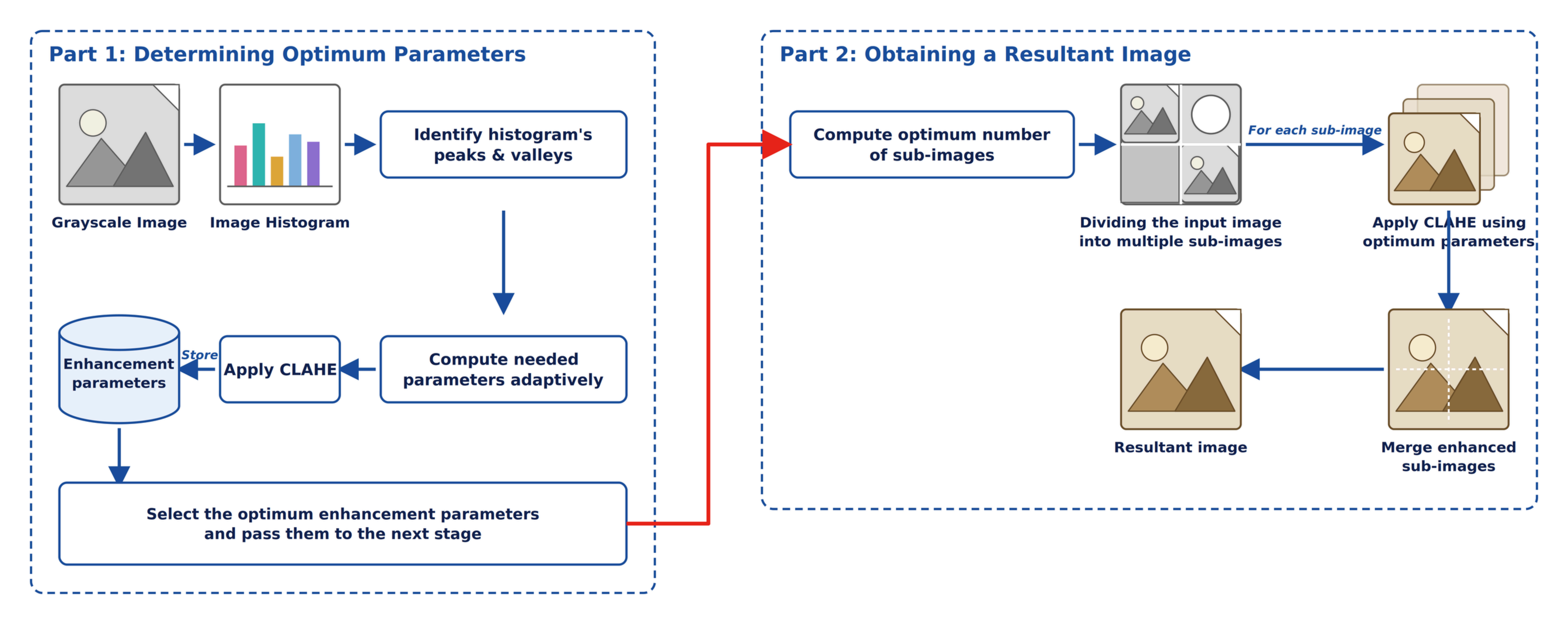
2

Local adaptive contrast enhancement

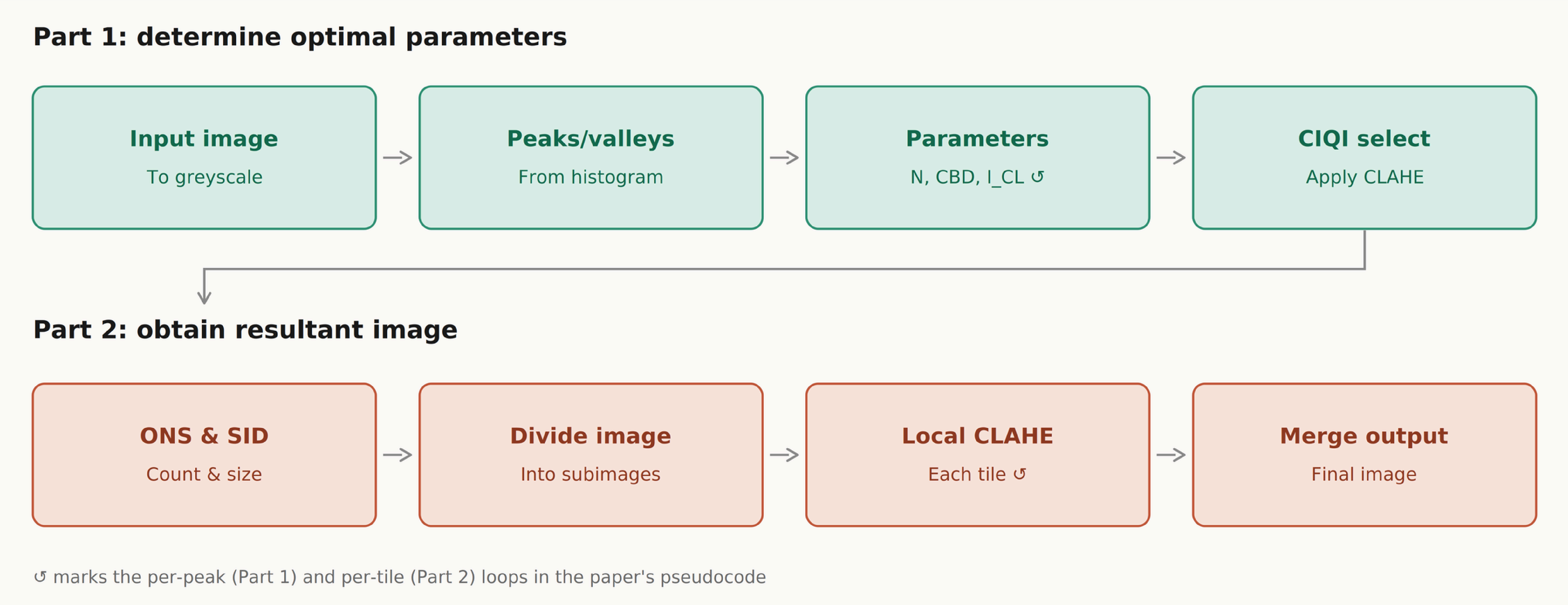
3

Outperforms 7 state-of-the-art methods

Block Diagram of Proposed CLALHE



Workflow of Proposed CLAHE



Key Terms: Peaks and Valleys

- Peak: a grey level whose pixel frequency is higher than both its immediate neighbours
- Valley: a grey level whose pixel frequency is lower than both its immediate neighbours
- NPeaks: total count of identified peaks in the histogram
- NValleys: total count of identified valleys in the histogram
- Peaks mark dominant intensities; valleys mark sparse/transition intensities

Gray levels	Frequency (Pixels	Peak	Valley	Gray levels	Frequency (Pixels	Peak	Valley
0	8087			11	2990		
1	3591			12	2391		
2	3329		✓	13	1952		
3	3508			14	1749		✓
4	4220			15	1794	✓	
5	5325			16	1702		✓
6	6283			17	1733		
7	6302	✓		18	1854		
8	5882			19	1930		
9	5156			20	2101		

Note: The symbol (✓) denotes the identified peaks and valleys.

Necessary Equations

N-parameter

```
N = sum(Valley[i]) / N_valleys
```

Average pixel frequency across identified valleys — keeps the clip-limit calculation grounded.

CBD

```
CBD = ceil( ln(N_peaks) )
```

Context block dimension — sets the CLAHE tile grid size from histogram complexity.

I_CL

```
I_CL = (Peak[i] + N) / max(Valley)
```

Candidate clip limit for each peak — low value, gentle enhancement; high value, aggressive.

CIQI

```
CIQI = (PSNR + Entropy) / AMBE
```

Fitness function — scores every candidate; highest CIQI is selected as "optimal."

Necessary Equations

Optimum Number of Subimages (ONS)

$ONS = \lceil \log_2(NValleys) \rceil$, rounded up to even if odd

number of subimages the image will be split into

Subimage Dimensions (SID)

- If ONS = 2: Height = M, Width = N/2
- If ONS > 2: Height = M/2, Width = N/(ONS/2)

exact pixel size of each subimage tile before independent CLAHE is applied and the tiles are merged

Reported Results

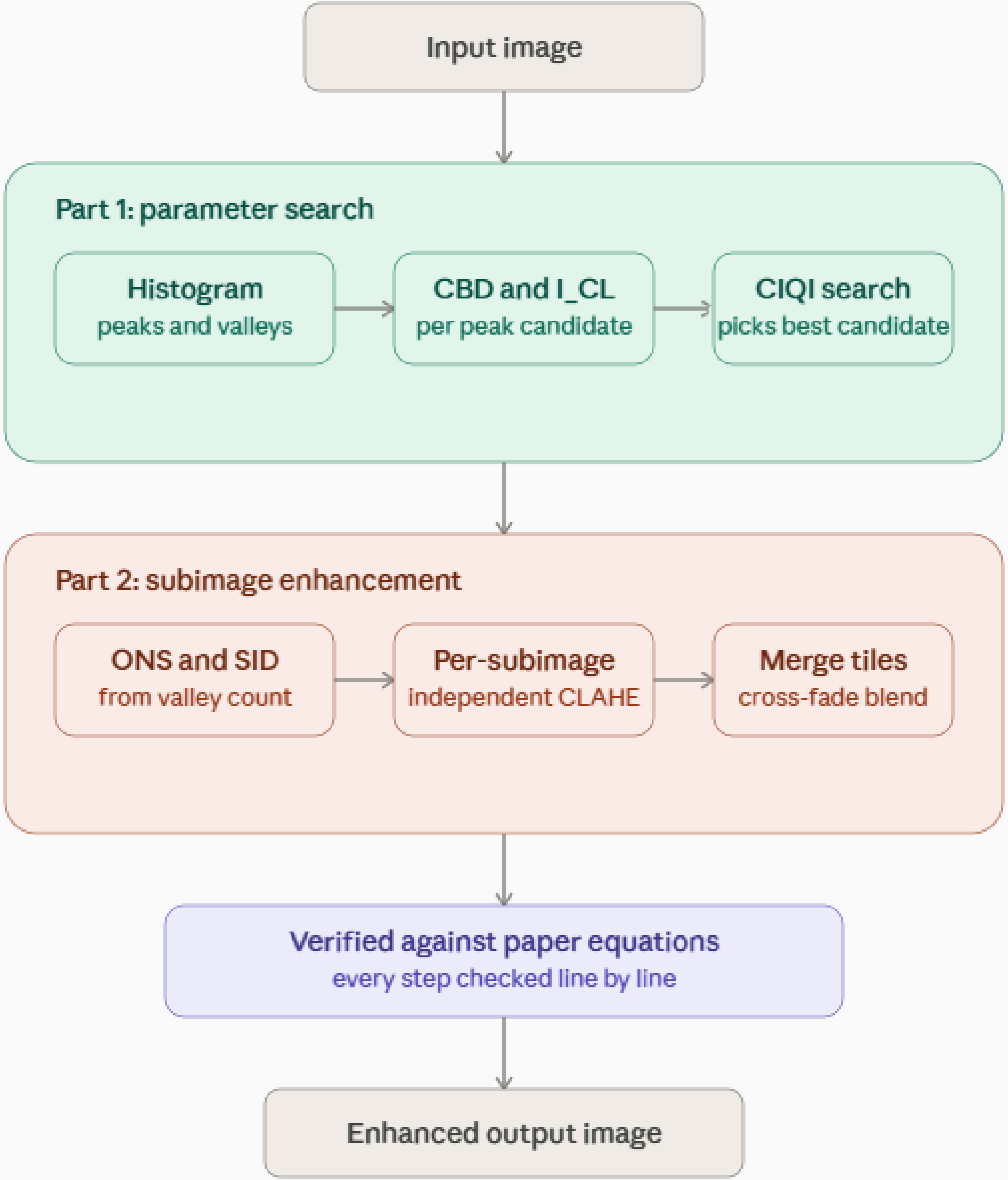
validated on 821 images across 3 datasets

- Faces 1999
- Pasadena-Houses 2000
- DIARETDB1

FIGURE 12. Resultant image of human face produced by the (a) original image, (b) CLAHE, (c) POSHE, (d) BHM, (e) IAECHE, (f) AEIHE, (g) Auto-CLAHE, (h) ACL-CLAHE and (i) the proposed CLALHE.

Our Implementation Process

- Python + OpenCV + SciPy
- Part 1: histogram → peaks/valleys → CBD & I_CL → CIQI search
- Part 2: ONS/SID → per-subimage CLAHE → cross-fade merge
- Verified against the paper's equations at every step



Block Artifact Issue

Input image

CLALHE output according to paper

Our Solution

Cross Fade Blending

- Read each subimage with an added margin of context beyond its border
- Composite overlapping subimages using a linear cross-fade instead of a hard cut
- Blend weights sum to exactly 1 at every shared boundary
- No new parameters, no change to the core algorithm — ONS, SID, CBD, I_CL all untouched
- Only the write-back/merge step changed

Before and After

CLALHE - paper

CLALHE blended

Domain Testing Setup

We tested CLALHE outside its original datasets, examining its performance on diverse domains including

- 🫁 Chest X-rays
- 👁️ Retinal/fundus images
- 🌊 Underwater images
- 🌙 Low-light images
- 📷 Normal/natural photographs

This evaluation involved 146 images and compared five different enhancement methods to assess adaptability and effectiveness.

Domain Specific Testing - Chest X-Ray

Input HE CLAHE CLALHE-paper CLALHE blended

Domain Specific Testing - Retinal Fundus

Input HE CLAHE CLALHE-paper CLALHE blended

Domain Specific Testing - Under Water

Input

HE

CLAHE

CLALHE-paper

CLALHE blended

Ground truth

Domain Specific Testing - Low Light

Input

HE

CLAHE

CLALHE-paper

CLALHE blended

Ground truth

Domain Specific Performace - Chest X-Ray

METHOD	PSNR	ENTROPY	AMBE	SSI	CII	RMSE
input (unmodified)	100.00	7.28	0	1.00	1.00	0
HE	18.97	6.98	11.29	0.8541	1.73	30.19
CLAHE	21.98	7.45	3.83	0.8416	1.48	20.41
CLALHE-paper	22.60	7.41	2.59	0.7981	1.45	21.63
CLALHE-blended	26.58	7.32	2.44	0.9191	1.26	12.11

The paper's six metrics, averaged over the domain. PSNR, SSI and RMSE are measured against the input, so they reward changing it least.

METHOD	FLAT-REGION SD IN	FLAT-REGION SD OUT	NOISE GAIN	CLIPPED %	SEAM STRENGTH	RUNTIME S
input (unmodified)	21.66	21.66	1.00	5.79	1.30	0.0001
HE	21.66	13.82	0.6891	6.03	1.25	0.0046
CLAHE	21.66	24.67	1.16	0.0136	1.26	0.0029
CLALHE-paper	21.66	25.26	1.19	0.0230	4.29	0.1234
CLALHE-blended	21.66	22.23	1.03	0.0129	1.28	0.1636

Domain Specific Performace - Retinal Fundus

METHOD	PSNR	ENTROPY	AMBE	SSI	CII	RMSE
input (unmodified)	100.00	6.07	0	1.00	1.00	0
HE	11.93	6.93	47.74	0.6891	1.27	66.19
CLAHE	23.38	6.33	14.00	0.7847	1.58	17.37
CLALHE-paper	28.01	6.13	7.98	0.8640	1.41	10.18
CLALHE-blended	27.72	6.17	8.51	0.8693	1.31	10.54

The paper's six metrics, averaged over the domain. PSNR, SSI and RMSE are measured against the input, so they reward changing it least.

METHOD	FLAT-REGION SD IN	FLAT-REGION SD OUT	NOISE GAIN	CLIPPED %	SEAM STRENGTH	RUNTIME S
input (unmodified)	7.78	7.78	1.00	11.82	1.14	0.0006
HE	7.78	14.13	2.31	11.81	1.01	0.0067
CLAHE	7.78	11.46	1.57	0	1.13	0.0081
CLALHE-paper	7.78	9.37	1.24	0	3.95	0.8629
CLALHE-blended	7.78	9.47	1.26	0	1.15	0.9925

Domain Specific Performace - Low Light

METHOD	PSNR	ENTROPY	AMBE	SSI	CII	RMSE
input (unmodified)	100.00	4.76	0	1.00	1.00	0
HE	5.95	5.29	112.38	0.0976	0.9534	128.72
CLAHE	23.97	5.55	14.46	0.6805	0.8662	16.53
CLALHE-paper	28.34	5.22	8.91	0.8144	0.8465	10.04
CLALHE-blended	28.22	5.21	9.08	0.8144	0.8435	10.15

The paper's six metrics, averaged over the domain. PSNR, SSI and RMSE are measured against the input, so they reward changing it least.

METHOD	FLAT-REGION SD IN	FLAT-REGION SD OUT	NOISE GAIN	CLIPPED %	SEAM STRENGTH	RUNTIME S
input (unmodified)	1.91	1.91	1.00	2.34	1.14	0.0001
HE	1.91	19.18	13.63	2.33	1.08	0.0023
CLAHE	1.91	3.72	2.10	0.0002	1.10	0.0023
CLALHE-paper	1.91	2.93	1.62	0.0002	1.36	0.0342
CLALHE-blended	1.91	2.95	1.62	0.0002	1.12	0.0466

Performance Against Ground Truth: Underwater and Low-Light Domains

METHOD	Δ PSNR VS TRUTH	Δ SSIM VS TRUTH	IMAGES WON
underwater — UIEB (n=40)			
HE	+0.75	+0.036	20/40
CLAHE	+0.99	+0.081	10/40
CLALHE-paper	+0.62	+0.061	2/40
CLALHE-blended	**+0.99**	+0.078	8/40
low-light — LOLv1 (n=15)			
HE	**+6.48**	+0.268	12/15
CLAHE	+1.36	+0.196	3/15
CLALHE-paper	+0.81	+0.125	0/15
CLALHE-blended	+0.83	+0.128	0/15

Summary on Diverse Domain Testing

Chest X-ray	HURTS	Seams cut across the lung fields; parameters are frozen across all 40 images.
Retinal fundus	HURTS	Same seam, over the retina — the paper's own showcase domain.
Underwater	MIXED	A real gain — but it only ties stock CLAHE, never beats it.
Low-light	NO-OP	Wins 0 of 15 images. Plain global HE gains 8× more against ground truth.
Natural photos	MIXED	The paper's home ground — a mild, safe enhancement, still seam-ridden.

Where CLAHE Breaks Down

1. **Visible seams**

   Creates brightness boundaries **2.8–4.3× stronger** than normal image detail

2. **Parameters that dont really adapt**

   Clip limit hits its minimum in **73% of low-light** and **55% of fundus** images . All 40 X-rays received the same parameters.

3. **Weak results in low light**

   CLAHE wins **0/15** low-light images. HE: +6.5 dB vs CLAHE: +0.8 dB

4. **Real problems on medical images**

   Seams can cross **important diagnostic regions** in X-rays and retinal images

5. **Can't handle real medical scan data**

   Designed for **8-bit images**; with 16-bit X-rays, it reads only **4.3% of pixels**

THANK YOU!